We would assume, G, H, K groups, $N \trianglelefteq G$ unless otherwise specified. Let's review some basics:

Theorem 1. *For all $G, H \leqslant K$ groups, $|GH| = \dfrac{|G| \cdot |H|}{|G \cap H|}$.*

Theorem 2 (Correspondence Theorem)**.** *If we have the canonical map $\varphi : G \to G/N$, then*

$$\varphi^* : \{\textit{Normal subgroup of } G \textbf{ containing } N\} \to \{\textit{Normal Subgroups of } G/H\},$$

$\varphi^*(N) = \varphi(N)$ is a bijection.

1 Actions and Sylow Theorems

Definition 1. An **action** of a group G on a set A is a function $\rho : G \times A \to A$ such that for all $g, h \in G, a \in A$:

$$\rho(e_G, a) = a, \rho(gh, a) = \rho(g, \rho(h, a)).$$

Remember the notation is also $\rho(g, a) := ga$. Now the set

$$Orb_G(a) := \{ga | a \in G\} \qquad Stab_G(a) := \{g \in G | ga = a\} \leqslant G$$

are called the **orbit** and **stabilizer** of an element $a \in A$ in the action.

Theorem 3. *The orbits forms an equivalence class on G. Moreover, $|Orb_G(a)| = [G : Stab_G(a)]$.*

Definition 2. If there is only one orbit, the action is **transitive**. If $Stab_G(a) = \{e_G\}$ for all $a \in A$, then the action is **faithful**.

Now we focus on the action $G \times G \to G : \rho(g, a) = gag^{-1}$ the conjugation action.

Definition 3. The **conjugacy classes** and the **centraliser** are the orbit, stabilizer of thie action respectively. Moreover, the set

$$\bigcap_{x \in G} Stab(x) = Z(G), \qquad \bigcap_{x \in S} Stab(x) = C_G(S)$$

are called the **center** of G and the **centralizer** of S in G respectively.

Theorem 4. *If $H \leqslant Z(G)$ and G/H is cyclic, then G is abelian.*

Theorem 5 (Class Number Formula)**.** *If a_i runs through the representative of centralizer such that $[G : C_G(a_i)] \geq 2$:*

$$|G| = |Z(G)| + \sum_i [G : C_G(a_i)]$$

Theorem 6 (Orbit Counting)**.** *If a finite group G acts on a finite set X, and define $Fix(g) = \#\{x \in X | gx = g\}$:*

$$\ell = \frac{1}{|G|} \sum_{g \in G} Fix(g)$$

where ℓ is the number of orbits in the action.

For what follows p, q are primes.

Theorem 7 (Cauchy)**.** *If p divide $|G|$ then there exists $g \in G$ such that g is order p.*

Definition 4. G is called a **p-group** if every element have prime powers order. Equivalently for finite group, the order of the group is a prime power.

Definition 5. If $|G| = mp^r$ (with $gcd(m, p) = 1$), a **sylow p-subgroup** of G is a group $H \leqslant G$ such that $|H| = p^r$.

Theorem 8 (Sylow)**.** *If $|G| = mp^r$ (with $gcd(m, p) = 1$)*

1. *G has at least one p-sylow subgroup, and every p-subgroup of G is also a subgroup of a sylow p-group.*
2. *The number of sylow sylow p-subgroup: n_p have the relation $n_p \equiv 1 (mod\ p)$ and $n_p | m$.*
3. *All sylow p-subgroups are conjugate to each other. In particular:*
   (a) *G acts on the set of sylow p-subgroup by conjugation: $gP := gPg^{-1}$. This action is transitive.*
   (b) *If $n_p = 1, m > 1$, it immediately follows that G is NOT simple.*

2 Series of Normal Groups

Consider a series of normal groups: $1 = G_1 \trianglelefteq G_2 \cdots \trianglelefteq G_r = G$:

Definition 6. This series alone is called a **subnormal** series. Moreover:

- The series is called **normal** if $G_i \trianglelefteq G$ for all $1 \leq i \leq r$.
- The series is called **composition series** if G_{i+1}/G_i are simple for all $1 \leq i \leq r$.

G_{i+1}/G_i are called the **factor groups** (at G_i).

Definition 7. Note that if we don't have a composition series, then for some $G_i \trianglelefteq G_{i+1}$ we can find a group $H \neq G_i, G_{i+1}$ and form a series

$$1 = G_1 \trianglelefteq G_2 \cdots G_i \trianglelefteq H \trianglelefteq G_{i+1} \cdots \trianglelefteq G_r = G$$

This is called a **refinement** of the composition series. It follows:

Theorem 9. *Any subnormal series of a finite group have a refinement to a composition series.*

Turns out that is the best that we can do:

Theorem 10 (Schreier)**.** *Any two subnormal series of subgroups of a given group have **equivalent** refinements to a composition series. Meaning any two series:*

- *Always have the same length r, and*
- *Their factors groups are isomorphic up to rearrangements.*

The result depends on a few lemma:

Lemma.

- *If $K \trianglelefteq H \leq G, f : G \to G'$ a homomorphism, then $f(K) \trianglelefteq f(H)$.*
- *If $K \trianglelefteq, H \leq G, N \trianglelefteq G$, then $NK \trianglelefteq NH$.*
- *If $A, B, C, D \leq G, A \trianglelefteq B, C \trianglelefteq D$, then*
  - $A(B \cap C) \trianglelefteq A(B \cap D)$, $C(D \cap A) \trianglelefteq C(D \cap B)$.
  - $[A(B \cap C)] \cap B \cap D = [C(D \cap A)] \cap D \cap B$

Theorem 11 (Butterfly Lemma)**.** *If $A, B, C, D \leq G, A \trianglelefteq B, C \trianglelefteq D$, then* $\dfrac{A(B \cap D)}{A(B \cap C)} \cong \dfrac{C(D \cap B)}{C(D \cap A)}$.

Proof of Schreier Refinement Theorem. Given two subnormal series $1 = H_0 \trianglelefteq H_1 \cdots \trianglelefteq H_r = G$ and $1 = K_0 \trianglelefteq K_1 \cdots \trianglelefteq K_s = G$, define:

$$H_{ij} = H_i(H_{i+1} \cap K_j), \qquad K_{ij} = K_j(K_{j+1} \cap H_i)$$

then the two series can be refined as ($r = 2, s = 3$, note that both refinement have length 6.)

$$1 = H_{00} \trianglelefteq H_{01} \trianglelefteq H_{02} \trianglelefteq \underline{H_{03} = H_{10}} \trianglelefteq H_{11} \trianglelefteq H_{12} \trianglelefteq H_{13} = G$$
$$1 = K_{00} \trianglelefteq K_{10} \trianglelefteq \underline{K_{20} = K_{01}} \trianglelefteq K_{11} \trianglelefteq \underline{K_{21} = K_{02}} \trianglelefteq K_{12} \trianglelefteq K_{22} = G$$

Note how the H, K are embedded in the refinement. And the existence and normal are by the butterfly lemma.

3 Solvable, Nilpotence, Products

Definition 8. A group H is a **characteristic subgroup** of G if for all $f \in \mathrm{Aut}(G)$ we have $f(H) = H$.

Definition 9. The **commutator subgroup** of G, denoted as G' is the subgroup generated by all the commutators $[x, y] = xyx^{-1}y^{-1}$.
We can then find the commutator subgroup of G', then the commutator subgroup of it...
define $G^{(r)}$ by the recurrent process $G^{(0)} = G, G^{(r+1)} = (G^{(r)})'$.

Definition 10. A group G is **solvable** if there exists r such that $G^{(r)} = 1$

We now collect all the properties we know about G' and solvability.

Theorem 12 (About G')**.**
- *G' is characteristic in G. In particular, it is normal in G.*
- *G is abelian $\Leftrightarrow$ G' is trivial. Moreover, G/G' is abelian.*
- *For any normal N, G/N is abelian $\Leftrightarrow$ $G' \leq N$, that is, G' is the minimal one to have this property.*

Directly from the definition, abelian groups are solvable, and non-abelian simple groups are **NOT** solvable.

Definition 11. If G is solvable, the subnormal series series $1 = G^{(r)} \trianglelefteq G^{(r-1)} \trianglelefteq \cdots \trianglelefteq G^{(1)} \trianglelefteq G$ is called the **derived series** of G.

Lemma. *If H is characteristic in K, $K \trianglelefteq G$, then $H \trianglelefteq G$.*

Proof. Consider the homomorphism $\phi_g(x) = gxg^{-1}$. □

With induction of length in series we conclude:

Theorem 13. *The derived series is a normal series.*

Theorem 14 (Properties of Solvable Groups)**.**
- *The following are equivalent:*
  - *G is solvable.*
  - *G has a normal series with abelian factor groups.*
  - *G has a subnormal series with abelian factor group.*
  - *All of the factor groups in the composition series are isomorphic to $\mathbb{Z}/p\mathbb{Z}$, p prime.*
- *If G is solvable, then any subgroup $H \leq G$, any quotient group G/H is solvable.*
- *If there is $N \trianglelefteq G$ such that $N, G/N$ are solvable, then G is solvable.*

Definition 12. A **central series** is a subnormal series $1 = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_n = G$, where $G_{i+1}/G_i \leq Z(G/G_i)$, or equivalently $[G, G_{i+1}] \leq G_i$. If a group has a central series, it is **nilpotent**.

Theorem 15. *If the following requirements hold: $H \trianglelefteq G, K \trianglelefteq G, \quad HK = G, \quad H \cap K = \{e\}$ Then $H \times K \cong G$, this is called the (internal) **direct product**.*

Theorem 16 (Properties of Nilpotent Groups)**.**
- *If G is a finite p-group, then G is nilpotent.*
- *Nilpotent groups are solvable. Moreover, if H, K are nilpotent, then $H \times K$ is.*
- *(The Structure Theorem.) If G is finite, then the following is equivalent:*
  - *G is nilpotent.*
  - *G is a direct product of **all** the sylow p-subgroup, where p ranges over all (possible) primes,*
  - *Every sylow p-subgroup is normal in G.*

Definition 13. If the following requirements about G hold:

$$H \trianglelefteq G, K \leq G, \quad H \cap K = \{e\}, \quad \phi \text{ is a homomorphism } K \to \mathrm{Aut}(H)$$

Then define multiplication on $H \times K$ as: $(h_1, k_1)(h_2, k_2) := (h_1\phi_{k_1}(h_2), k_1k_2)$, this gives a group structure. And is called the (internal) **semidirect product** of H, K, denoted as $H \rtimes_\phi K$.